

The relationship between charge shape characteristics and fill level and lifter height for a SAG mill



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ABSTRACT

A predictive model that allows mill fill level and the lifter height to be determined from measurements of the head, bulk and impact toe positions for a generic SAG mill has been developed using data from a series of three dimension DEM simulations. A Design of Experiment (DOE) approach is used to define a series simulation conditions with the two factors being the lifter height and the fill level. The charge structure is characterised by the locations of the head, shoulder, bulk toe and impact toe. These are determined by visual analysis of the particle trajectories. The spatial variation of the liner pressure distribution can provide valuable information on the structure of the charge. This was found to vary systematically with lifter height and fill level. The relationships between the visual key locations and the key features of the pressure distributions were identified and found to be well described by linear equations. Finally, parametric models are fitted to the DEM data for both the lifter height and fill level allowing them to be estimated from measurements of the key charge locations. These could be used as part of a control strategy for the mill and/or to assist with re-line scheduling.

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1. Introduction

Tumbling mills are large cylindrical chambers that rotate about their central axis at a reasonable fraction of the speed required to centrifuge the particulate charge. This usually consists of a combination of grinding media (steel balls) and rocks (in various stages of size reduction). The rotation of the mill leads to shear between adjacent layers of particles. Grinding is produced by a range of mechanisms including high speed impact, abrasion, chipping and nipping of small particles between large ones (Delaney et al., 2013). SAG mills are the largest type tumbling mills used (with diameters up to 42 ft or 12.6 m) and are used for primary grinding, mainly in mineral processing, using a coarse feed (typically up to 250 mm diameter rock) coming from a crushing stage of the grinding circuit. Ball mills are somewhat smaller and use a larger fraction of grinding media with a finer feed size and are typically used for some form of second stage grinding.

The charge distribution within such mills usually has a kidney like shape with particles moving upwards near the mill liner, reaching a high point where the centrifugal and gravitational forces balance (usually called the “shoulder”) and then flowing

downwards with a free surface that is typically bi-linear with a steep gradient down from the shoulder followed by a shallower gradient out to the start of the charge (usually called the “toe”). If large and/or steep lifters are used in the liner design to lift the charge then some particles are thrown on high ballistic trajectories (in what is usually called the cataracting part of the flow) and impact on the liner in a region above the toe. The top of this direct liner impact region is often called the “impact toe”. The highest point of the liner that is still in contact with the particles is also called the “head”. The locations of these key characteristic locations of the charge provide significant information about the current operation of the mill and can be used in controlling the mill. These locations all vary with critical mill conditions such as the mill speed, the fill level and the conditions of the lifters.

A major measurement challenge has been and remains to be the reliable measurement of these locations. Measurement is possible in laboratory scale experiments, such as when using

- Perspex front mill models as measured by Ward (1992) and used in Cleary et al. (2003). Katubilwa and Moys (2011) used a Perspex mill to explore load behaviour in a tumbling mill and its relationship to charge volume slurry viscosity. Conduction based probes (Moys et al., 1996) although tend to be more useful for identifying the slurry position rather than

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the solid charge; and [Mulenga and Moys \(2014\)](#) used a proximity probe and a conductivity sensor to measure the position of the media charge and the slurry.

- PEPT (Positron Emission Particle Tracking) by: [Parker et al. \(1994\)](#), [Volkwyn et al. \(2011\)](#) and [Govendar et al. \(2013\)](#).

In full size mills, efforts to use vibration sensors based on accelerometers mounted on lifter bolts ([Spencer et al., 1999](#); [Campbell et al., 2001](#); [Behera et al., 2007](#)) and acoustic measurement ([Pax, 2001](#); [Huang et al., 2009](#)), have met with different amounts of success. Some success has also been reported by [Keshav et al. \(2011\)](#) using Sensomag developed by Magotteaux using an instrumented lifter in operational mills. [Tano et al. \(2005\)](#) used strain gauges embedded in lifters to monitor charge properties in tumbling mills. [Kalala et al. \(2005a\)](#) used an instrumented lifter bar to compare measured forces with DEM predictions.

The Discrete Element Method (DEM) has been used successfully for modelling many industrial applications, particularly grinding mills. The first such usage was report by [Mishra and Rajamani \(1992, 1994\)](#) and has then been used [Rajamani and Mishra \(1996\)](#), [Cleary \(1998, 2001a,b,c, 2004, 2009\)](#), [Cleary et al. \(2008\)](#), [Herbst and Nordell \(2001\)](#), [Morrison and Cleary \(2004, 2008\)](#), [Djordjevic \(2003, 2005\)](#), [Djordjevic et al. \(2006\)](#), [Jayasundara et al. \(2006\)](#), [Kalala et al. \(2005a,b\)](#), [Powell et al. \(2006, 2011\)](#), [Carvalho and Tavares \(2011\)](#), [Tavares and de Carvalho \(2009\)](#) and many authors since then. Conference special issues SAG2006 ([Mular et al., 2006](#)), DEM2007 ([Cleary and Morrison, 2008](#)) and SAG2011 ([Major et al., 2011](#)) provide a good selection of such DEM mill related papers. The current status of DEM usage in comminution is described by [Weerasekara et al. \(2013\)](#). DEM is able to predict the shoulder, toe and impact toe locations for specific mills and operating conditions and can be used to relate the measured charge structure back to the liner and operating conditions.

In this paper, we develop a predictive model using DEM that allows mill load (expressed as fractional fill) and the lifter wear state to be determined from bulk and impact toe, shoulder and head positions. A Design of Experiment (DOE) method is used to define a series of 3D DEM simulations that were performed on a generic SAG mill for lifter geometries in various stages of wear and for various fill levels. The characteristic charge structure locations used for this study are estimates for the head, shoulder, bulk toe and impact toe. These are determined by two different methods, the first involving visual analysis of the particle trajectories and the second using mean liner pressure distributions. The relationships between the key locations determined by the two approaches are evaluated. A parametric model is fitted to the DEM data for the key locations. This provides a fast method for estimating the mill liner wear state and charge fill level from measured key charge locations. These can then be used to determine the mill condition liner condition and current fill level from measurements of the charge location. This information can be used as part of a control strategy for the mill and to assist with re-line scheduling.

2. DEM method

DEM simulation involves following the motion of every particle in the flow and modelling each collision between combinations of particles and between particles and their environment such as the liner of a mill. The soft particle implementation used here is described in more detail in [Cleary \(1998, 2004, 2009\)](#). Briefly, the particles are allowed to overlap and the amount of overlap Δx , and normal v_n and tangential v_t relative velocities determine the collisional forces via a contact force law. A linear spring-dashpot

model is used for particle–particle and particle–boundary collisional force. See [Thornton et al. \(2013\)](#) for other contact models that could be used. Details of the models and comparisons for single particle oblique collisions with walls are given there for all such contact models of inelastic contacts.

The normal force:

$$F_n = -k_n \Delta x + C_n v_n \quad (1)$$

consists of a linear spring to provide the repulsive force and a dashpot to dissipate a specified proportion of the relative kinetic energy. The maximum overlap between particles is determined by the stiffness k_n of the spring in the normal direction. The normal damping coefficient C_n is chosen to give the required coefficient of restitution ε (defined as the ratio of the post-collisional to pre-collisional normal component of the relative velocity), and is given in [Thornton et al. \(2013\)](#).

The tangential force is given by:

$$F_t = \min \left\{ \mu F_n, \sum k_t v_t \Delta t + C_t v_t \right\}, \quad (2)$$

where the vector force F_t and velocity v_t are defined in the plane tangent to the surface at the contact point. The integral term represents an incremental spring that stores energy from the relative tangential motion and models the elastic tangential deformation of the contacting surfaces, while the dashpot dissipates energy from the tangential motion and models the tangential plastic deformation of the contact. The total tangential force F_t is limited by the Coulomb frictional limit μF_n , at which point the surface contact shears and the particles begin to slide over each other.

3. Generic SAG mill geometry and operating conditions

This current study was carried out using a series of DEM simulations of a 3D slice of a generic SAG mill. The “generic” SAG mill, detailed in [Table 1](#), was constructed to investigate the effect of lifter height/face angle (i.e. lifter wear state) and fill level on charge shape for a given rotational speed. The profiles used in this study for the lifter shape are shown in [Fig. 1](#). The different lifter heights can arise from either use of different size new lifters in the mill or as an approximate series of increasingly lower lifters that represent their basic wear cycle. These profile variations are inputs to the model and are not an adaptive prediction of the wear on the liners or the resulting evolution of the liner shape (as has previously been demonstrated by [Cleary, 2001b](#); [Cleary et al., 2010](#) or [Cleary and Owen, 2015](#)). The largest lifter had a height of 300 mm and a face angle of 15°; while the shallowest lifter had a height of 50 mm and a face angle of 25°. The lifter profiles with intermediate heights and face angles are shown in [Fig. 1](#). Other combinations of lifter height and face angle can be considered but for this initial study they are combined into one lifter profile variable that can be used as a single factor.

Using a reduced factorial, central composite DOE allows the principal effects, interactions and second order effects to be examined while minimizing the number of simulations. For a two parameter study, the central composite DOE requires five different

Table 1

Summary of the specifications of the slice model of the generic SAG mill used for the DEM simulations.

Shell inner diameter	8.4 m
Slice length	0.6 m (along the cylindrical belly)
Number of lifters	56
Liner configuration	High/high
Lifter base width	320 mm
Lifter height	50–300 mm
Lifter face angles	15–25° (varying with lifter height)

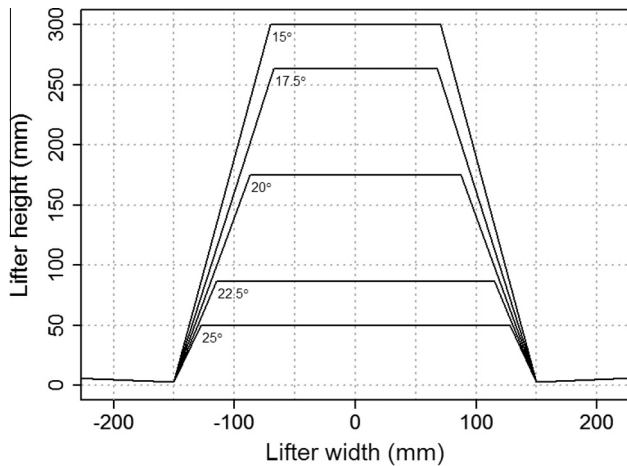


Fig. 1. Lifter profiles, ranging from a height of 300 mm, down to the shallowest one with a height of 50 mm. Each profile is annotated with its face angle.

volumetric fill levels and five different lifter wear states. Given the minimum and maximum values for each parameter the DOE method determines the optimal intermediate values. For this study, the following ranges were used:

- Fill levels ranged from 20% to 35% by volume representing the majority of the usual operating range for SAG mills.
- Lifter heights ranged from 50 mm to 300 mm with face angles suitable for each height.

The DOE method defines a series of twelve simulations which includes four centre-points repeats for error estimation. The details for each simulation are shown in Table 2. All twelve simulations were conducted using the same generic mill diameter (8.4 m) and the same rotational speed of 11 rpm (75% critical speed).

A representative SAG mill charge composition was used. The particle size distribution (PSD) of rocks and balls are given in Table 3. This had a rock top size of 200 mm and a smallest resolved rock size of 25 mm. The ball distribution ranged from 145 mm down to 30 mm. The same charge composition was used for all runs. In all cases, the density of ore particles and steel balls were 3000 and 7850 kg/m³, respectively. A constant ball charge of 6% by volume was used for all simulations. This means that when the overall fill level of the mill was changes for cases given in Table 2 this was achieved by varying the amount of rock in the charge. This corresponds to a mill operating at its required ball charge level while the rock level can be varied by the operator changing the feed rate or by the rock hardness or coarseness changing.

Table 2
Summary of the two-parameter DOE conditions for the DEM simulations.

Simulation case	Fill level (%)	Lifter height (mm)
1	27.5	50
2	27.5	175
3	32.8	263.4
4	35	175
5	27.5	175
6	27.5	300
7	27.5	175
8	22.2	263.4
9	27.5	175
10	20	175
11	22.2	86.6
12	32.8	86.6

Table 3

Particle size distribution (PSD) for the ore and steel ball components of the charge used for all DEM simulations.

Ore size (mm)	Ore PSD (%)	Ball size (mm)	Steel ball PSD (%)
–200 + 140	8	–145 + 100	30
–140 + 100	12	–100 + 60	35
–100 + 70	15	–60 + 45	20
–70 + 50	15	–45 + 30	15
–50 + 35	25		
–35 + 25	25		

The material properties and collision parameters for each combination of materials is given in Table 4. The coefficient of restitution is highest for steel–steel collisions which are relatively elastic and lowest for rock–rock collisions which have significant local fracture at contacts leading to low restitution. The friction coefficient is smallest at 0.4 for the ball–ball collisions and highest for the rock–rock ones since the rocks are typically quite rough. DEM simulation of mill behaviour has been shown to be only weakly sensitive to the specific choices of these material parameters (Cleary, 2001a). Since the aim here is to simulate a representative charge that is consistent with many mills, these mid-range values have been chosen. These are similar to those that have been used in previous DEM studies (Cleary et al., 2008; Delaney et al., 2013; Franke et al., 2015). The spring stiffness used was 20×10^6 N/m which was chosen to give an average overlap of the order of 0.8% of the average particle diameter. This is sufficient to give accurate mill simulations (see Cleary, 2004). The shear stiffness used was half that of the normal stiffness.

4. Impact toe, bulk toe, shoulder and head locations

4.1. Key feature locations identified visually

Visualisations of the charge shape and streak patterns were used to determine the impact toe, bulk toe, shoulder and head positions. Standard DEM particle visualisations, as shown in Fig. 2, were used to estimate the bulk toe and shoulder locations. Streak patterns, as shown in Fig. 3, were used to estimate the impact toe and head locations. A streak representation is formed by tracing the trajectory of each particle position over a specified time interval which in this case is 0.2 s. The streak pattern then represents the motion of the particle whereas a standard rendering of the particles shows only the current position. This makes it easier to identify the extent of the cataracting stream, the shape of the bulk charge and improves the accuracy of the identification of the shoulder. It is equivalent to what happens when a photograph is taken with a finite exposure time. This approach was used in Cleary et al. (2003) to match the visual appearance of the

Table 4
Material properties and collision parameters for the DEM simulations.

Material property/collision parameters	Values
Steel ball density (kg/m ³)	7850
Rock density (kg/m ³)	3000
Steel–steel collisions	
Coefficient of restitution	0.75
Friction coefficient	0.4
Steel–rock collisions	
Coefficient of restitution	0.5
Friction coefficient	0.5
Rock–rock collisions	
Coefficient of restitution	0.3
Friction coefficient	0.8

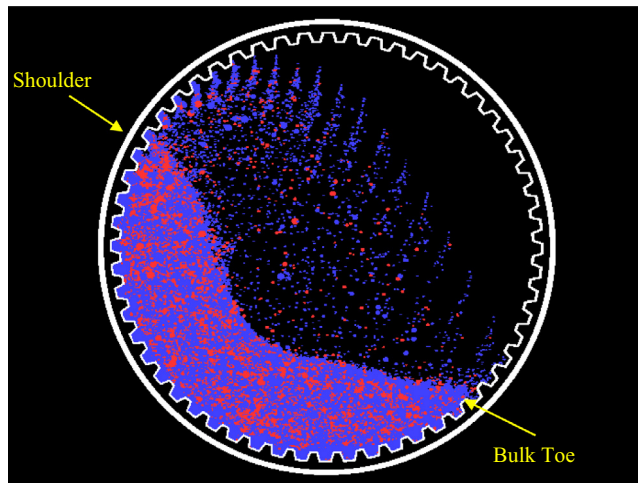


Fig. 2. Standard particle visualisation showing location of bulk toe and shoulder. The rocks are shown as blue and the balls as red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

experimental photographs used for comparison with the DEM predictions. The shoulder and bulk toe location are indicated on Fig. 2 while the head and the impact toe are shown on Fig. 3. In these figures the rocks are shown as red while the rock is shown as blue. To allow the angular locations of the key charge locations to be accurately determined an angular grid is projected over the streak plot (in Fig. 3).

Table 5 shows a summary of the impact toe, bulk toe, shoulder and head positions predicted for the twelve DEM simulation cases defined in Table 2. The estimates of the bulk toe and shoulder were obtained using a procedure based on one given by Cleary et al. (2003) where the shoulder is defined as the point where the trajectory of particles diverge from the mill, and the bulk toe is the highest position reached by the bulk of the charge in the toe region. The impact toe is the location where the highest trajectories in the cataracting stream intersect the liner. This is always above the bulk toe location. The head is the highest point of the liner that is still in contact with the particles. This is always above the shoulder

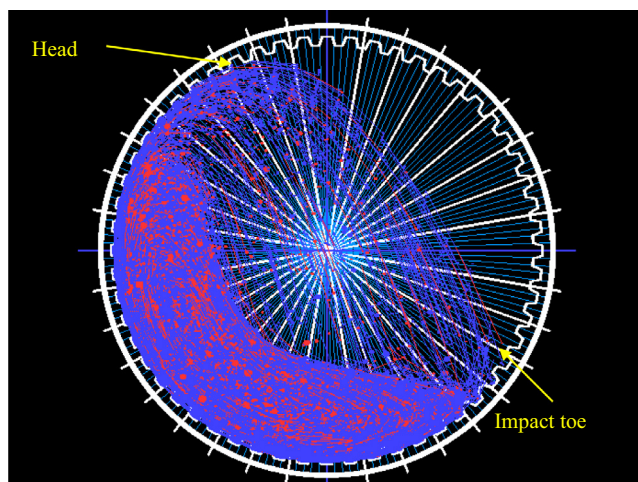


Fig. 3. Streak pattern showing location of impact toe and head. The rocks are shown as blue and the balls as red. An angular grid is superimposed to allow the key charge locations to be measured. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

Predicted locations of the bulk toe, impact toe, shoulder and head positions for the twelve 3D DEM simulation cases.

Case	Fill level (%)	Lifter height (mm)	Head (°)	Shoulder (°)	Bulk toe (°)	Impact toe (°)
1	27.5	50	130	150	318	306
2	27.5	175	110	148	314	330
3	32.8	263.4	104	146	318	334
4	35	175	110	146	320	330
5	27.5	175	110	152	314	330
6	27.5	300	98	152	312	338
7	27.5	175	111	152	314	332
8	22.2	263.4	104	158	306	336
9	27.5	175	112	152	314	332
10	20	175	112	164	306	330
11	22.2	86.6	123	160	310	316
12	32.8	86.6	122	148	324	320

position. The location of these features is given as an angle of anti-clockwise rotation from the 3 o'clock position.

Fig. 4 shows a representative streak snapshot of the steady charge distribution for each of the twelve operating condition modelled. There are four replicates of the centre point conditions in this DOE. These are Cases 2, 5, 7 and 9 and they are shown in the first four panels of the figure. The fine details of the charge differ but there is very close similarity in all the key features of the charge structure. The bulk toe is consistently located at 314°, while shoulder is in the range 148–152°. The impact toe is in the range 330–332°.

The variability between these cases arises from the manner of the construction of a DEM model of mill from the probability distributions used to specify the charge composition (Table 3). For each such simulation, particles are generated with their size and size class determined randomly based on the probability weightings of these size classes and the ratio of balls to rocks. The finer size classes have large numbers of particles and therefore have lower relative variability since this is inversely proportional to the square root of the number of particles in the class. For the larger size classes where there are typically much smaller numbers of particles and sometimes quite small numbers then the amount of variability can be quite significant. So although these repetitions are statistically equivalent (in the sense that they are created from the same probability distributions) the numbers and masses of particles in each rock and ball class can vary in non-trivial ways. In this case, the number of rock particles in the coarsest class in each of the four repetitions is 103, 116, 100 and 101 respectively. This represents a 7% coefficient of variation around the mean of 105. In contrast the variability in the smallest rock class is only 0.6% around the mean of 61,838 particles. Usually, in using DEM to model industrial particle processes, such questions about the reproducibility of the results which are subject to the variability of the particle sets generated are ignored. These issues are also quite analogous to ones relating to the variability of feed to real mills where the feed from varies instant to instant from the average size distribution that is usually used to characterise the feed. The repeated simulations arising from the DOE here provides an opportunity to examine the degree to which this variability in the charge composition affects the qualitative and quantitative features of the charge. The results found here show that the key charge locations are insensitive to this variability in the charge composition and that any one DEM instance of a mill configuration is sufficient to adequately predict the charge structure.

The difference between the location of the impact toe and the bulk toe is in the range of 16–18°. This indicates the extent of liner locations that are subject to direct liner impacts. From the streak pictures in Fig. 4, it is clear that only a very small fraction of the

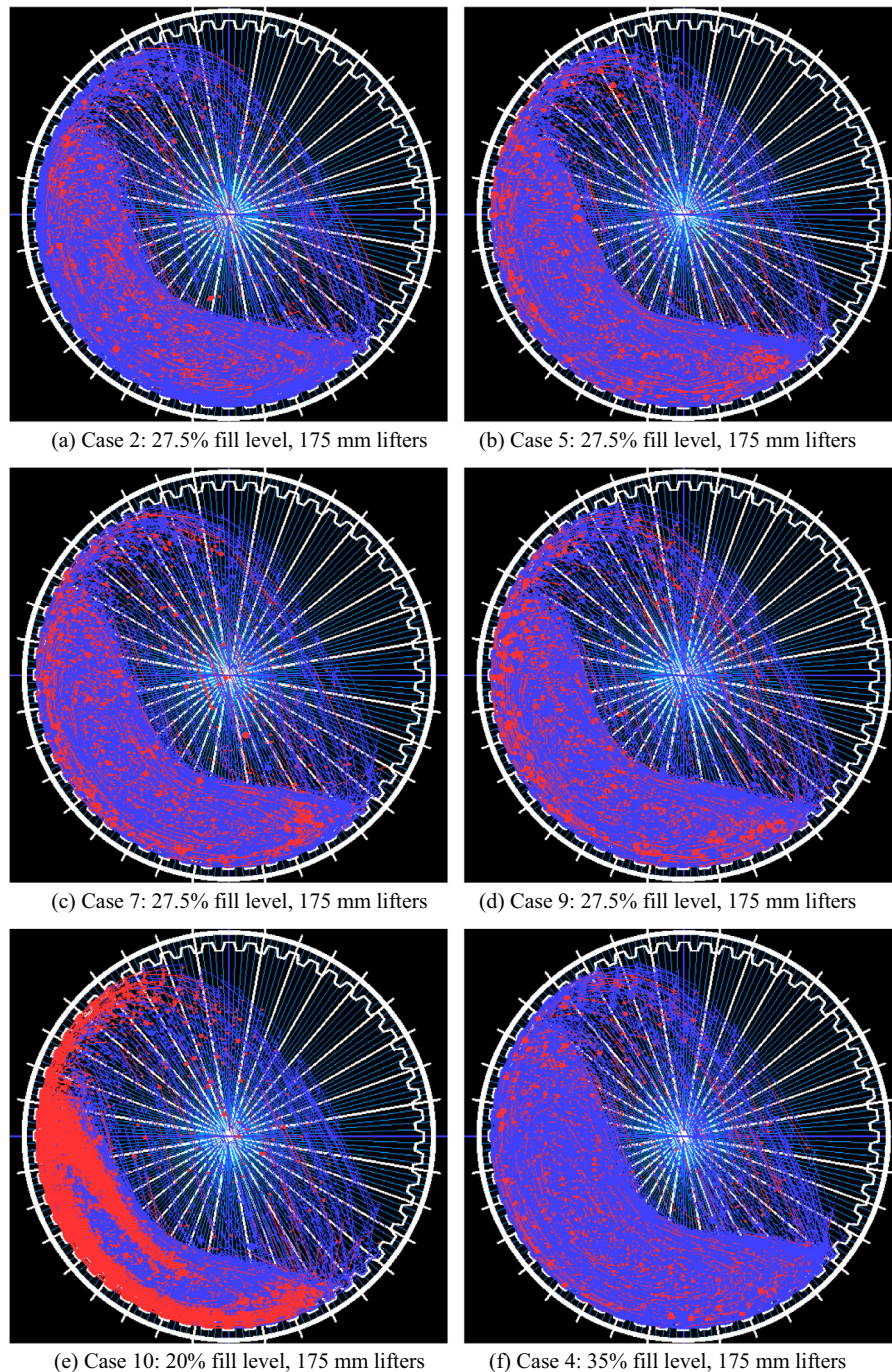


Fig. 4. Steady state charge behaviour shown as streaks for the twelve cases simulated. The rocks are shown as blue and the balls as red. An angular grid is superimposed to allow the key charge locations to be measured. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

particles that directly impact the liner are media (shown as red) with the dominant fraction of the cataracting stream being rock particles. This is the result of radial segregation generated by the centrifugal forces that causes smaller particles (typically rocks) to migrate outwards while larger particles (typically balls) migrate away from the liner towards the middle of the charge.

Case 10 (bottom left panel of Fig. 3) has a 20% fill level. The ball fraction is constant at 6% so the amount of rock is sharply lower than for the earlier panels. This causes the strong relative visual increase in the red component of the charge. The lower fill level clearly leads to a lower bulk toe but has limited effect on the impact toe which is predominantly controlled by the heights of

the lifters. This means that for lower fill levels a larger fraction of the charge is in the cataracting stream and experiences direct liner impacts. The shoulder position is also necessarily lower. In contrast, a higher fill level such as the 35% fill shown in the bottom right panel of figure leads to a higher shoulder and bulk toe but only limited effects on the head and impact toe locations. The dominance of rock in the upper half of the cataracting stream due to the radial segregation is clearly visible.

The effect of lower and higher lifter heights (which could result from higher and lower wear on the lifters respectively) is shown in the second page of Fig. 4 with the shallower lifters in the left column and the higher lifters in the right column. The strongest effect of the

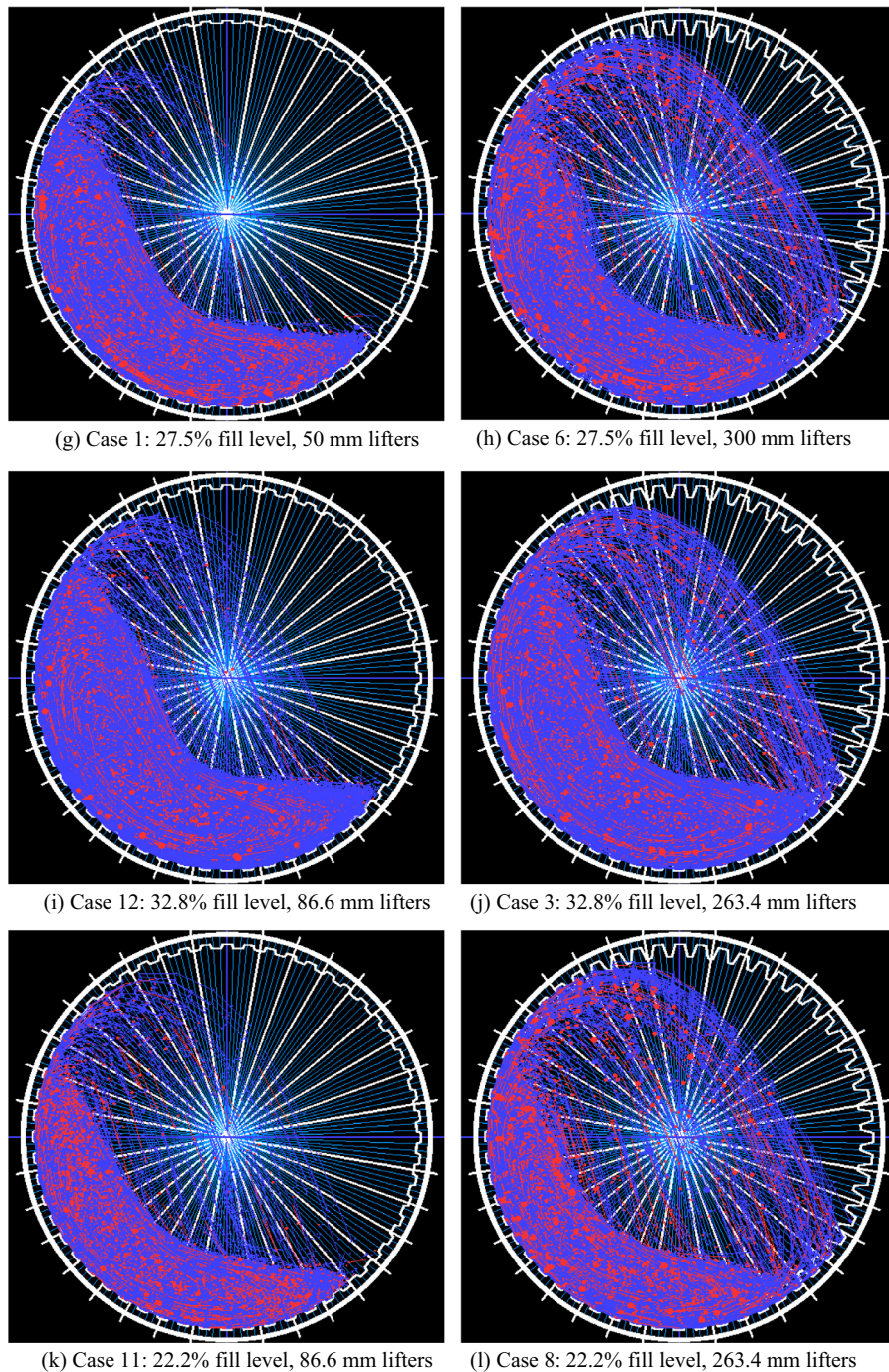


Fig. 4 (continued)

lifter height is on the extent and location of the cataracting stream. For higher lifters, a greater fraction of the charge is thrown higher leading to a higher impact toe. For the lower lifters the cataracting stream is significantly reduced in size and in the amount of charge that is thrown through the air. An important aspect of this is that there are on a few direct liner impacts and the impact toe is close to being aligned with the bulk toe. As the lifters become very low (representing a significantly worn state) there is also a slight slumping of the bulk of the charge. This is most noticeable when comparing Case 6 (300 mm lifters) with Case 1 (50 mm lifters) where the charge is located around 6° lower in the mill.

Figs. 5–12 summarise the quantitative variations of each of the key charge locations with the control variables of fill level and lifter

wear state. Fig. 5 shows the variation of the head location as a function of fill level for the three different lifter heights (wear states). There is only a weak variation with fill level, but the head location is clearly a strong function of the height of the lifters. Newer lifters are able to lift the head of the charge much higher than is the case for worn lifters. The sensitivity of the fill level dependence of the head position increases slowly as the lifters become more worn – with little dependence observed for higher lifters but a reasonable of variation for worn lifters.

Fig. 6 shows the location of the shoulder plotted against the fill level for three different lifter heights. The charge shoulder varies little with changing lifter height over the range considered. In contrast, the shoulder is strongly dependent on the fill level, moving

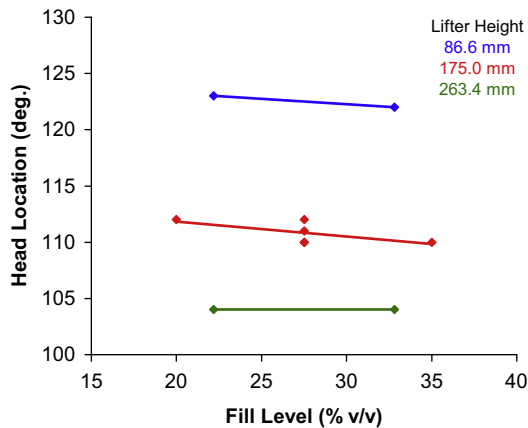


Fig. 5. Location of the head of the charge versus fill level.

linearly higher in the mill with increasing fill level (which corresponds to a numerically smaller shoulder angle). The fill level dependence of the shoulder is the same for all the lifter heights.

Fig. 7 shows the location of the bulk toe for varying fill level for three different lifter heights. There is a strong linear dependence with the bulk toe moving higher as the charge fill level increases. The bulk toe also increases with decreasing lifter height as the smaller lifters are less able to slow the cascading flow of particles down to the toe. The dependence of the bulk toe on lifter height is weaker than its dependence on the fill level.

Fig. 8 shows the location of the impact toe as function of the fill level for three different lifter heights. The fill level has only a weak influence the position of impact toe and the nature of this dependence differs with lifter height. For larger lifters, the impact toe angle decreases slightly with fill level. For intermediate lifters, the impact toe is invariant with fill level, but for worn lifters the impact toe increases with fill level.

Fig. 9 shows the variation of the head location as a function of the lifter wear state for three different fill levels. It changes approximately linearly with the height of the lifters, with newer lifters producing much higher head positions (smaller angular locations) than worn lifters. The insensitivity to fill level is shown by the close proximity of the three curves. Fig. 10 shows that the location of the shoulder varies little with lifter height but is higher with higher for higher fill levels as indicated by the decreasing shoulder angles. Fig. 11 shows that the bulk toe location increases as the lifters wear for all the fill levels. The bulk toe also increases with fill level.

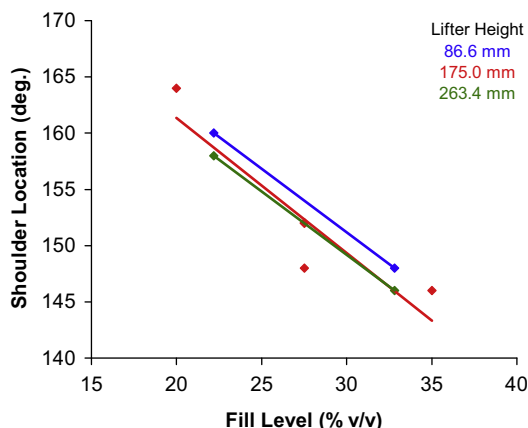


Fig. 6. Location of the shoulder of the charge versus fill level.

Fig. 12 shows that the angular position of the impact toe increases approximately linearly related with the lifter height. The impact toe for newer lifters is positioned higher than the impact toe for worn lifters. The fill level of the mill has little influence on the position of impact toe.

For a fixed lifter height:

- Increasing the volume of material in a mill produces a larger charge which results in both the shoulder and the bulk toe positions moving to higher locations around the internal surface of the mill.
- The head of the charge appears to be invariant to the fill level and the impact toe is only slightly affected by changes in the fill level. These parameters are clearly affected by lifter height.

For fixed fill levels:

- Declining lifter height (due to increasing wear) reduces the height of the head and the height of the impact toe, due to reductions in the amount of cataracting material and the height of the ballistic trajectories in this part of the flow.
- The shoulder and the bulk toe are only slightly sensitive to changes in lifter height.

The significant relationships identified from these results are the:

- Bulk toe height is dependent on the fill level.
- Shoulder height is dependent on the fill level.
- Impact toe height is dependent on the lifter height.
- Head height is dependent on the lifter height.

4.2. Correlation between operating conditions and key feature locations

A correlation analysis was performed to quantify these relationships between the operating conditions (lifter height and fill level) and the key locations in the charge. Several correlations are identified and given in Table 7. The most important are:

- The head and impact toe both correlate very strongly with the lifter height. The lifter height is most sensitive to changes in the head with a correlation of -0.98 meaning that the lifter height decreases linearly with increasing head angle (which corresponds to a decreasing height of the head in the mill). The lifter height dependence on the impact toe is positive (in that they increase together) and nearly as strong.

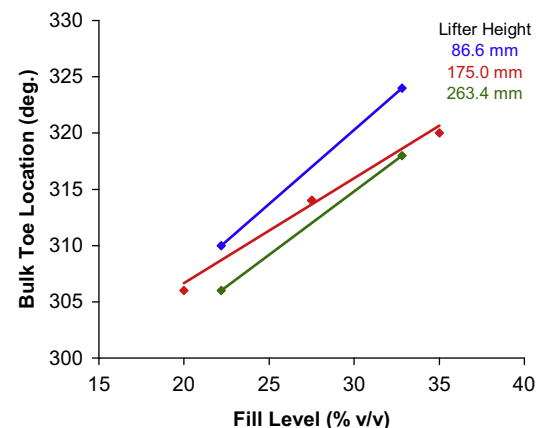


Fig. 7. Location of the bulk toe of the charge versus fill level.

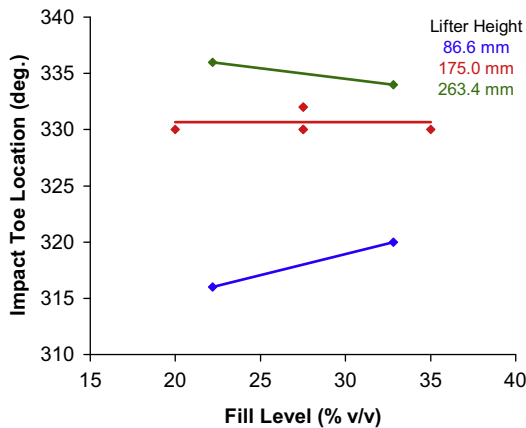


Fig. 8. Location of the impact toe of the charge versus fill level.

- The shoulder position and the location of the bulk toe are strongly correlated to the fill level with correlation factors of -0.94 and 0.92 respectively. So the fill level is equally dependent on both the shoulder and bulk toe angles, decreasing with increasing shoulder angle and decreasing with decreasing bulk toe angles.

These are consistent with the visual findings in the previous section. Correlations of the lifter height dependence on the shoulder and bulk toe were too weak to be statistically significant. There was no statistically significant dependence of the fill level on the head and impact toe locations. This means that the dependencies between the operating conditions and charge structure are simple and strong.

4.3. Key feature locations identified from mean liner pressure distribution

The key charge positions, impact toe, bulk toe, shoulder and head can also be determined, in principle, directly from the level of liner impact forces predicted by the DEM simulation. Using this type of approach is somewhat analogous to some of the proposed experimental charge measurement techniques based on liner vibrations and acoustic emissions. An important question is then about the degree to which the key locations arising from the liner pressure correlate with the visually determined ones.

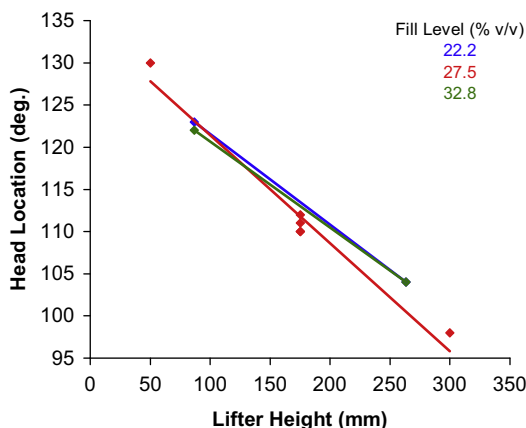


Fig. 9. Location of the head of the charge versus lifter height.

For each simulation the time averaged spatial distribution of the force applied to the liner by the charge is calculated as a function of angular location in the mill. These are normalised by area to give pressure distributions. Fig. 13 shows an example of the pressure distribution which corresponds to Case 2. In the uppermost part of the charge (around 12 o'clock in Fig. 4a) the mean pressure is zero since there is no charge in this region making contact with the liner. Moving in an anti-clockwise direction, the pressure begins to rise slowly in the region of the head (label A in Fig. 13) and then much more strongly in the region of the shoulder where the gradient of the pressure distribution curve increases (label B). The pressure rises nearly linearly from the inflexion point (B) reaching a peak (C) at around 240° . It then decreases to a local minimum (labelled D) at around 300° . This is followed by a very sharp rise around the bulk toe region to a peak at around 310° (E) which is followed by a sharp decline to zero pressure (label F) as the impact toe is reached. These locations will be used in the analysis of the variation of the liner pressure with operating conditions which are shown in Fig. 14 for all twelve DEM simulation cases. The angular position is again measured in terms of an anti-clockwise rotation from the 3 o'clock position. The horizontal axis on each graph ranges from 90° (12 o'clock) to 360° (3 o'clock).

Whilst the pressure distributions in Fig. 14 vary sensitivity with both the lifter height and the fill level they can all be interpreted as a superposition of two characteristics sub-distributions. The first characteristic is associated with forces due to pressure the bulk of the charge which is pressed against the liner and the second which is associated with the cataracting stream impacting on the toe region and directly on the liner in the region between the bulk toe and the impact toe (typically at angles above 280°). The bulk charge distribution is roughly Gaussian but is skewed to the right reflecting the centre of the charge being closer to the toe than the shoulder. The structure of the second distribution varies strongly with the conditions varying from nearly Gaussian for high fill and low lifter height cases to having noisy sharp multi-peaks when the cataracting stream is strong.

The top four panels of Fig. 14 are replicates of the mid-point conditions and show a very high degree of similarity (as should be expected if the pressures are well averaged over time). For Case 10 (bottom left panel) the low fill level causes the cataracting stream to contain a relatively larger proportion of the charge, the toe region peak increases significantly in size relative to the main distribution on the left. This causes location C to become just an inflexion point rather than a local maximum and eliminates the decline and local minimum labelled as D. The right side of the toe distribution a number of sharp high frequency oscillations super-imposed resulting from high intensity impacts on the liner.

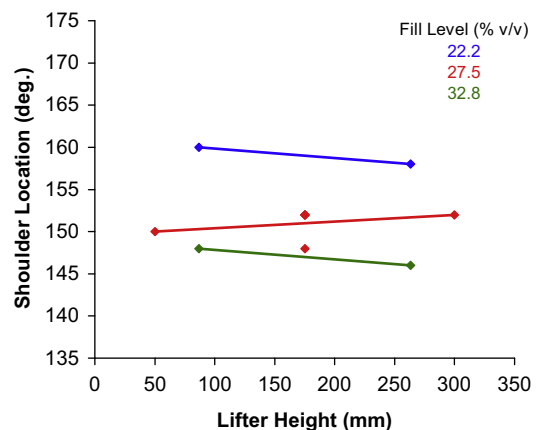


Fig. 10. Location of the shoulder of the charge versus lifter height.

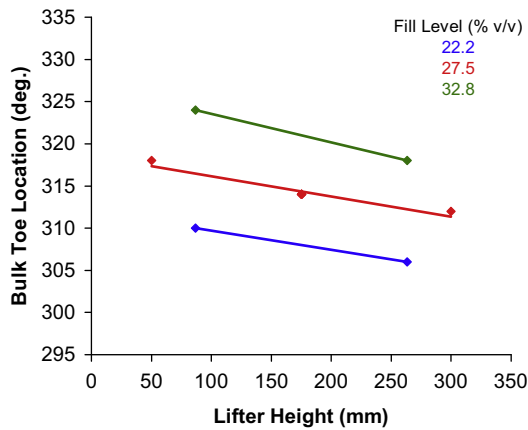


Fig. 11. Location of the bulk toe of the charge versus lifter height.

Further time averaging would smooth out these minor peaks. An increased fill level (bottom right panel) has the opposite effect with the toe region peak on the right being strongly reduced relative to the main charge distribution. The effect of advanced wear has an even more marked effect on the magnitude of the toe region distribution with this becoming very weak for all the fill levels. In contrast, the effect of high lifter heights significantly increases the relative contribution of the toe region distribution and causes it to have significant high frequency structure corresponding to the high intensity direct liner impacts. The magnitude of the variations in the toe region distribution with liner height and fill level indicate that significant and exploitable variations in noise and vibration should occur that could be used as the basis of successful measurement of the toe region of the charge.

4.4. Correlation between visual and mean liner pressure key locations

We choose four of the key points on the pressure distributions (in Fig. 14) that appear to correlate to a useful degree with the four key positions that we have used to describe the shape of the charge (namely the head, shoulder, bulk toe and impact toe). The angle at which the mean pressure curve becomes non-zero can be related to the head of the charge. We associate the position of the bulk toe with the angular position of the highest mean liner pressure. The impact toe is associated (by definition) with the angle at which the mean pressure becomes zero again. The position of the shoulder is more difficult to relate to the structure of the pressure distribution. In most of the cases, the mean pressure rises slowly when

moving from the head region towards the shoulder. There appears to be an identifiable change in gradient of the mean pressure causing an upward kink in the pressure curves. On this basis, we use the angular position of this kink as an indicator of the shoulder location. The suitability of these associations will be tested later by examining the degree of correlation with the visible positions of the key charge locations. Table 6 gives the location of the four characteristic points in the pressure distributions for all the twelve cases in this study. The pressure rise location varies from 95° for a high lifter and high fill level to 133° for a moderate fill level and a shallow lifter. The pressure kink location varies from a low of 145° for high fill and low lifter up to 158° for a low fill level and low lifter height. The maximum pressure location varies from 304° for a low fill and intermediate lifter height up to 326° for an intermediate fill level and high lifter. The pressure fall location varies from 330° for a low fill and low lifter up to 356° for intermediate fill level and lifter height.

Correlation factors were calculated to determine the degree to which the features used to characterise the mean pressure distributions are related to key locations that describe the charge (namely the impact toe, bulk toe, shoulder and head). The correlations from all pair-wise comparisons are summarised in Table 8.

The head and impact toe both correlate strongly with the “pressure rise” and “pressure fall” points. There is a moderate correlation between the head and the pressure maximum and a weak correlation between the impact toe and the maximum pressure. There is no dependence of the head or the impact toe on the kink location.

The shoulder and bulk toe both correlate well with the “kink” in the pressure curve. The shoulder has a strong positive correlation with the pressure “kink” while the bulk toe has a strong negative correlation with the pressure “kink”. The shoulder location also correlates moderately with the pressure maximum location. The shoulder position does not depend on either the rise or the fall positions of the pressure. The bulk toe shows no dependence on any of the rise, maximum or fall locations – it varies only with the kink location.

Table 8 shows that none of the key charge locations correlate well with the pressure maximum. As indicated earlier, the pressure distributions appear to be made up from the superposition of two characteristic component pressure curves. The maximum of the pressure curve is sometimes part of the second characteristic curve which is likely to be associated with the cataracting stream impacting in the bulk toe/impact toe region but sometimes arises from the first characteristic curve which is associated with the weight of the bulk of the charge. Depending on the fill level and lifter height, the maximum pressure could arise from the peak of

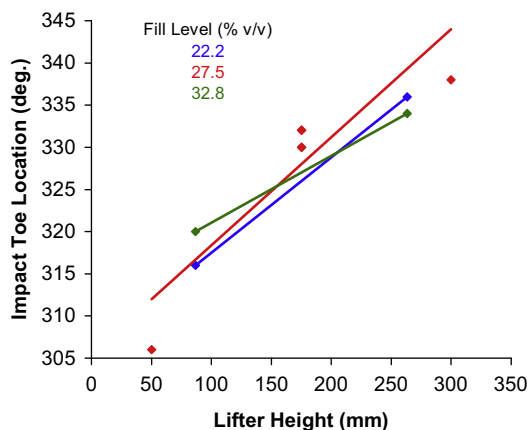


Fig. 12. Location of the impact toe of the charge versus lifter height.

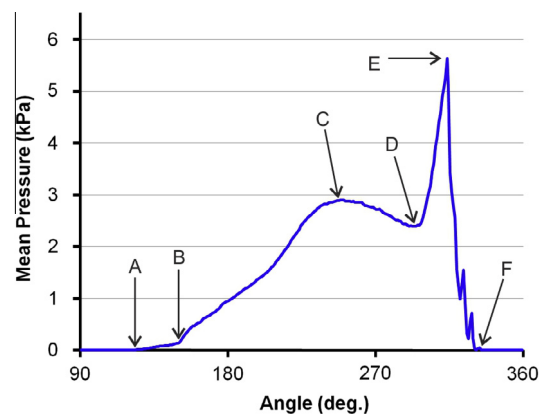


Fig. 13. Location of the key locations in the liner pressure distribution.

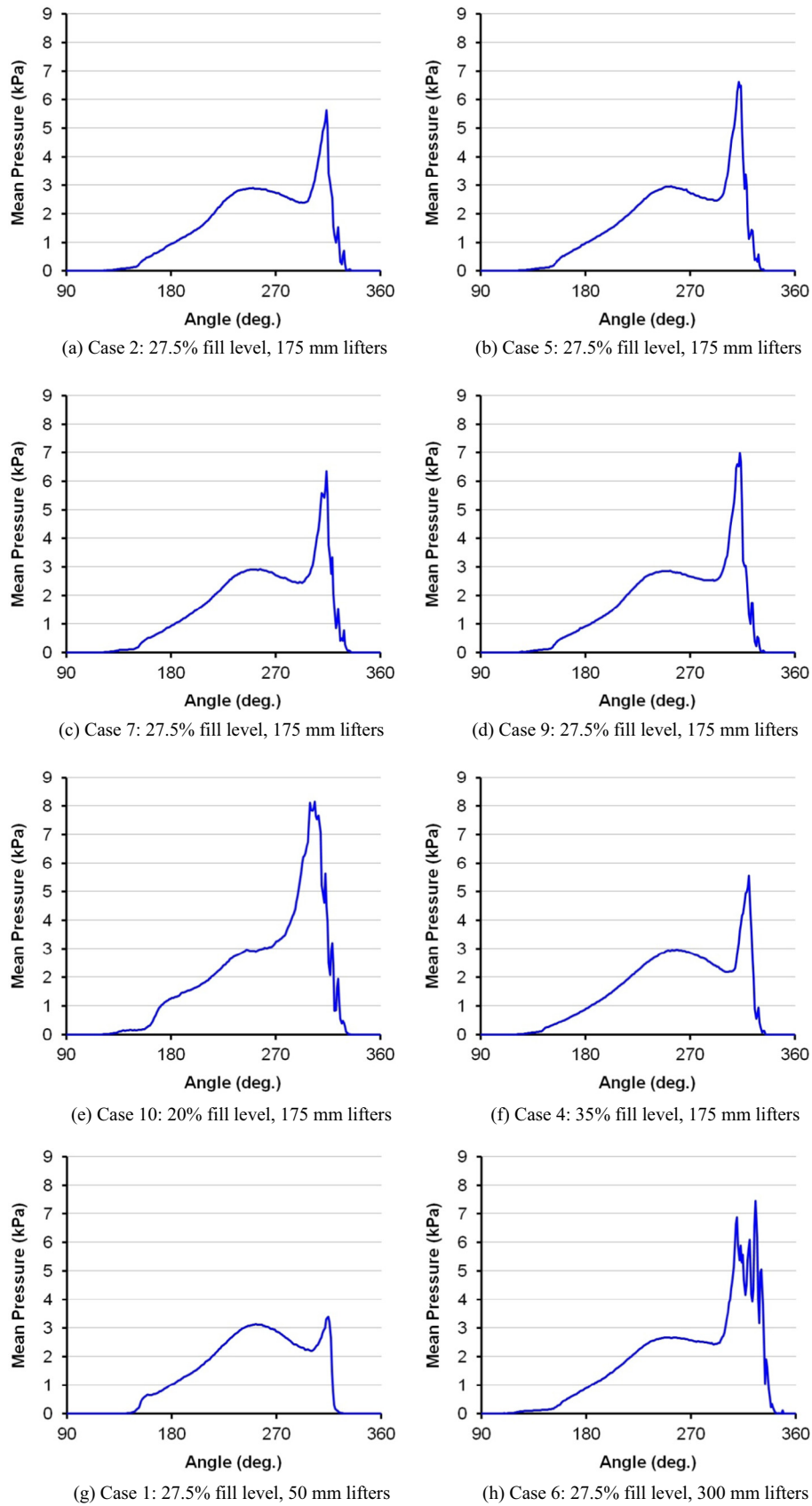


Fig. 14. Predicted liner pressure distribution as a function of angular position around the mill (3 o'clock = 0°, increasing angle is anti-clockwise).

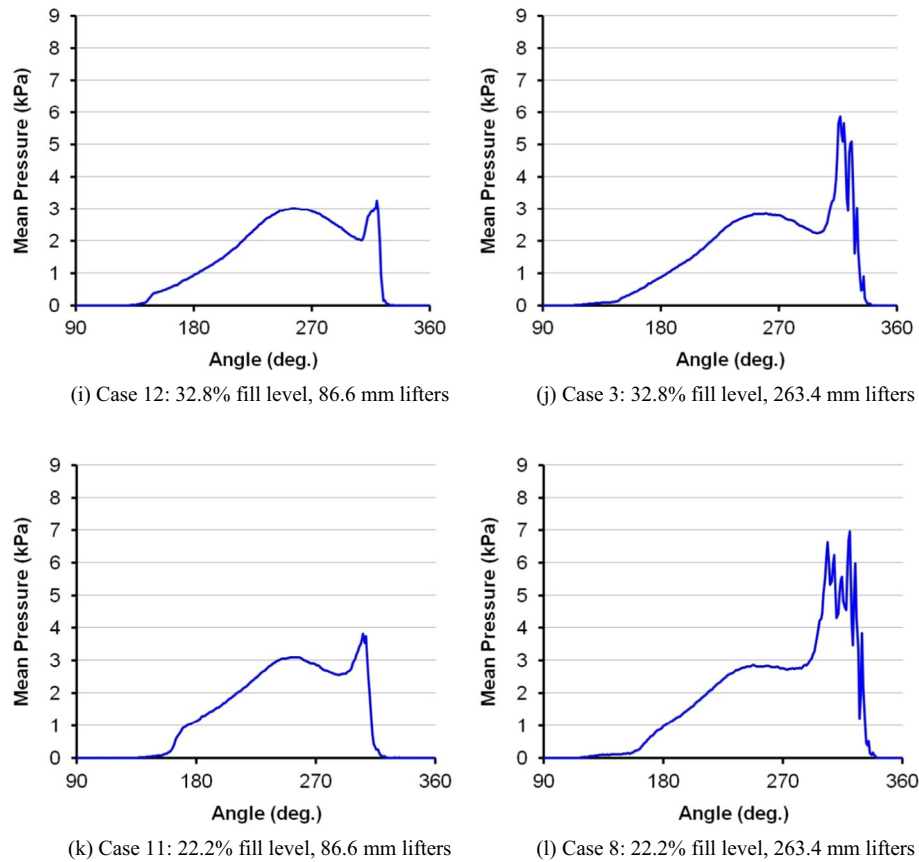


Fig. 14 (continued)

Table 6

Angular location of key features from the mean liner pressure distributions.

Case	Fill level (%)	Lifter height (mm)	Pressure rise (°)	Pressure kink (°)	Pressure max. (°)	Pressure fall (°)
1	27.5	50	133	153	315	328
2	27.5	175	112	151	314	335
3	32.8	263.4	95	147	317	343
4	35	175	108	143	321	340
5	27.5	175	110	151	312	356
6	27.5	300	104	152	326	346
7	27.5	175	115	150	314	335
8	22.2	263.4	101	157	321	342
9	27.5	175	114	151	313	335
10	20	175	114	157	304	334
11	22.2	86.6	128	158	306	330
12	32.8	86.6	123	145	320	337

Table 7

Operating conditions and key locations correlation factors.

Operating conditions	Key location	Correlation factor
Lifter height	Head	−0.98
Lifter height	Shoulder	−0.12
Lifter height	Bulk toe	−0.20
Lifter height	Impact toe	0.91
Fill level	Head	−0.20
Fill level	Shoulder	−0.94
Fill level	Bulk toe	0.92
Fill level	Impact toe	0.14

Table 8

Key locations and pressure curve features correlation factors.

Key location	Pressure curve feature	Correlation factor
Head	Rise	0.93
Head	Kink	0.27
Head	Maximum	−0.57
Head	Fall	−0.94
Shoulder	Rise	0.27
Shoulder	Kink	0.88
Shoulder	Maximum	−0.57
Shoulder	Fall	0.28
Bulk toe	Rise	0.03
Bulk toe	Kink	−0.86
Bulk toe	Maximum	0.33
Bulk toe	Fall	−0.01
Impact toe	Rise	−0.89
Impact toe	Kink	−0.26
Impact toe	Maximum	0.41
Impact toe	Fall	0.83

either component curve. This means that the maximum pressure is not a consistent indicator of the dynamic behaviour as its origin depends on the operating conditions. This leads to poor correlations to the charge structure.

As previously identified, the location of the shoulder (S) has a strong positive correlation with the pressure “kink” (K). Fitting a line of best fit gives the following relationship:

$$S = 1.0526 \times K - 6.35, \quad (3)$$

with a reasonable residual of $R^2 = 0.77$.

The head (H) has a strong positive correlation with the location of both the pressure rise (R) and fall (F) features in the pressure curve. The rise is the more intuitive of the two for use in creating a predictive model. The line of best fit of the rise and the head, with a good residual of $R^2 = 0.87$, is:

$$H = 0.7657 \times R - 25.96. \quad (4)$$

The impact toe (I) has a strong positive correlation the location of both the pressure rise (R) and fall (F) features. The fall is the more intuitive of the two for use in creating a predictive model. The line of best fit of the fall and the impact toe is:

$$I = 1.4078 \times F - 145.53, \quad (5)$$

which has a reasonable residual of $R^2 = 0.64$.

The bulk toe (B) has a strong negative correlation with pressure kink (K). The best linear fit is:

$$B = 392.1 - 0.7682 \times K, \quad (6)$$

with a reasonable residual of $R^2 = 0.77$.

The correlations shown in Tables 7 and 8 and the four fitted linear Eqs. ((3)–(6)) indicate that the key locations of the charge structure in a tumbling mill can be robustly determined from liner pressure measurements. The question of how to best measure the pressure distribution on the liner remains an open one with a range of possible technologies having been explored (as reviewed in the introduction). If these measurements can be successfully made then this model can be used to determine the structure of the charge. In the next section, we develop a model to predict fill level and the height of the lifters from known values of these key charge locations.

5. Predictive model for fill level and lifter height using key charge locations

A predictive model for fill level and the lifter height, using known values of the key charge locations (head, shoulder, bulk toe and impact toe), was developed using multiple linear regression analysis applied to the DEM predicted charge structure information in Table 5. It is simply the calculation of the estimated value of a dependent variable y_{est} from a linear combination of independent variables, x_i , for $i = 1, 2, 3, \dots$, or, in other words, a sum of linear terms including the error:

$$y_{est} = a_0 + a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_kx_k + \dots + error. \quad (7)$$

The a_i parameters are the regression coefficients for the respective independent variables. Quadratic terms, including cross-terms, were included as additional independent variables (e.g. $x_kx_j = x_i, x_j$). Two predictive models were developed here, one for the fill level F and one for the lifter height L . The independent variables used in the two models are: the location of the head H , the shoulder S ,

the bulk toe B , and the impact toe I . A number of multiple regression analyses were carried to identify the best predictive models for estimating lifter height and fill level from knowledge of the key locations. The models that were considered, the most significant parameter combinations found for each model and their fitting residuals are given in Table 9.

The model that provides the best prediction of the fill level is:

$$F = a_0 + a_1H + a_2B + a_3BH, \quad (8)$$

where the regression coefficients are: $a_0 = -600.37$, $a_1 = 3.094$, $a_2 = 2.067$, $a_3 = -0.01045$. The predictions of the fill level using this second order model are shown in Fig. 15 plotted against the actual fill level (red diamonds) from the original DEM data in Table 5. The solid black line represents the line for predicted values equal to the actual values. This shows that the second order model is an excellent predictor of fill level given the location of the head and bulk toe are known. Fig. 16 is a surface plot of the fill level predicted by the above regression model (8) showing its shape and the nature of its dependencies on the head and bulk toe locations. The fill level is lowest when the head value H is larger (meaning that the head is lower in the mill) and when the bulk toe angle is smaller (meaning that the bulk toe is also lower in the mill). As the head and bulk toe both rise in the mill then the fill level causing this is higher. The highest fill level occurs for the highest bulk toe location at 324° and the highest head location which occurs for the smallest head angle of 98° . From this model, the fill level can be determined using just the head location and the bulk toe. These need to be found either visually or indirectly from liner force information via the earlier model in Eqs. (4) and (6).

The model that provides the best prediction of the lifter height is:

$$L = b_0 + b_1H + b_2I + b_3IH, \quad (9)$$

where b_i are the regression coefficients: $b_0 = -6745.5$, $b_1 = 53.47$, $b_2 = 23.38$, and $b_3 = -0.184$. The predictions of the lifter height using this second order model are shown in Fig. 17 against the actual lifter height (red diamonds) from the original DEM data in Table 5. The solid black line represents the line for predicted values equal to the actual values. This shows that the second order model is a very good predictor but slightly underestimates the lifter height. Fig. 18 is a surface plot of the lifter height predicted using the above regression model (9) showing its shape and the nature of its dependencies on the impact toe and head positions. The lifter height is greater when the head angle is smaller (that is the head is

Table 9
Best predictive model obtained from multiple regression analysis.

Operating conditions	Regression type	Significant parameters	R^2
Fill level	Linear	Head, bulk toe	0.96
Fill level	Linear with interaction	Head, bulk toe, head $\times$ bulk toe	0.97
Fill level	Full quadratic	Head, bulk toe, head $\times$ bulk toe	0.97
Lifter height	Linear	Head, impact toe	0.96
Lifter height	Linear with interaction	Head, impact toe, head $\times$ impact toe	0.98
Lifter height	Full quadratic	Head, impact toe, head $\times$ impact toe	0.98

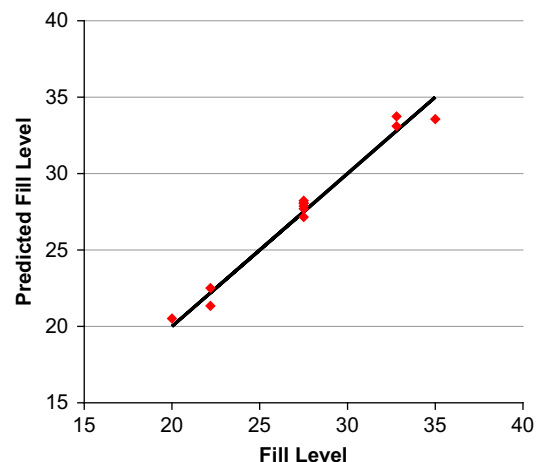


Fig. 15. Plot of predicted fill level versus actual fill level.

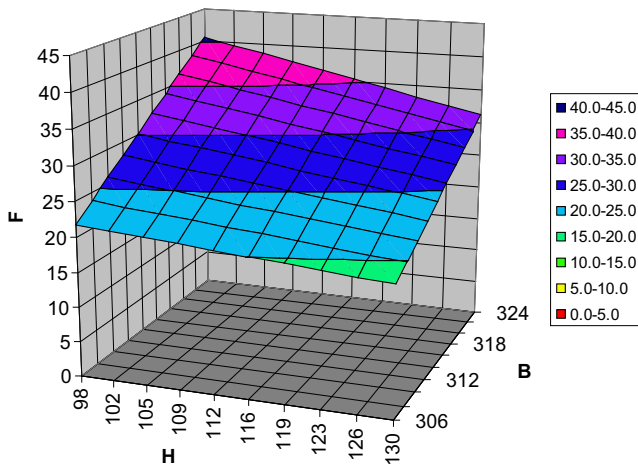


Fig. 16. Surface plot of predicted fill level (F) versus head (H) and bulk toe (B).

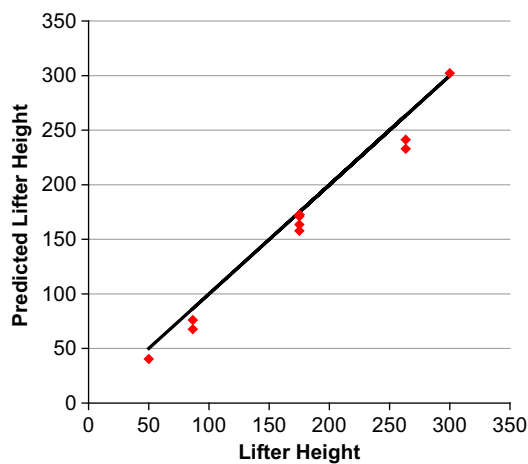


Fig. 17. Plot of predicted lifter height versus actual lifter height.

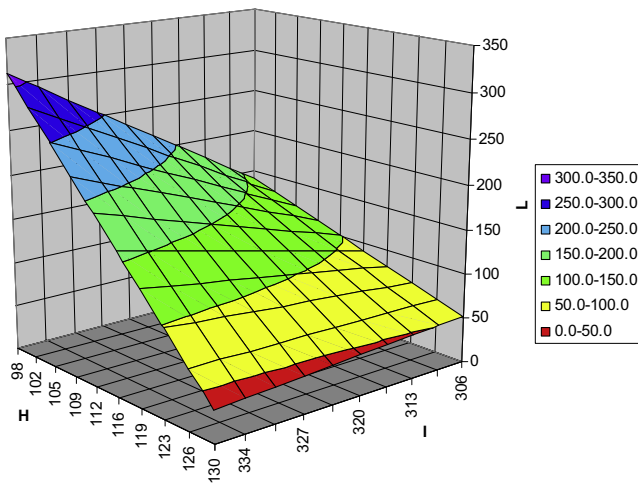


Fig. 18. Surface plot of the predicted lifter height (L) versus head (H) and impact toe (I).

higher in the mill) and decreases linearly with the head with the smallest lifter height occurring for the lowest head position (largest head angle). The gradient of this dependency is largest when the

impact toe is largest and decreases as the impact toe angle decreases. The difference in lifter height predicted between the highest and lowest head positions is 250 mm for the highest impact toe location (334°) but decreases to less than 100 mm difference when the impact toe is lower (at 306°).

These predictive models (8) and (9) allow mill load and lifter height to be determined directly from just three of the key locations, namely: bulk toe, impact toe and head positions. It is interesting to note that the location of the shoulder is not required for either of the predictive models. This is quite important since the shoulder is often poorly defined and hard to identify visually, so not needing an accurate estimate of the shoulder is positive for determining liner and operating conditions from the charge shape. These predictive regression models can be potentially used in monitoring processes to relate measured charge outputs (such as head, bulk toe and impact toe) to the current operating and liner wear conditions.

6. Conclusions

Three dimensional DEM simulations that were performed for a generic 8.4 m diameter SAG mill for a range of lifter heights and fill levels. A Design of Experiment (DOE) approach was used to define the levels of these factors and the combinations required to create a series of eight sets of simulation conditions. The charge structure was characterised by identifying the locations of the head, shoulder, bulk toe and impact toe using visual analysis of the particle trajectories. The variation of the key charge locations with the operating condition factors was quantified.

The spatial variation of the steady state pressure of the charge against the liner provides a second approach to characterising the charge behaviour. The liner pressure varies in a systematic manner with variations in the mill operating conditions. These can be characterised using four key locations and the variations of these locations were used to quantify the pressure variations. Correlation analysis showed that there were clear relationships between the visual shape of charge (head, shoulder, bulk toe and impact toe), the key features obtained from the pressure distributions. Linear models were sufficient to describe these dependencies which can be used to determine the charge structure from any measured liner pressure distributions.

The main relationships found are:

- The fill level correlates strongly with the bulk toe and shoulder positions. These positions also correlate strongly with the kink in the pressure curve.
- The lifter height correlates strongly with the head and impact toe positions and these positions correlate well with the force rise and fall features from the liner pressure distributions.

Multiple linear regression analyses of the DEM generated charge structure data for the set of DOE simulation conditions provided excellent predictive models for both the lifter height and fill level. These provide a fast method for estimating the mill liner height and charge fill level from measured estimates of the key charge locations or from measured liner pressure distributions. This information could be used as part of a control strategy for the mill and could assist with decision making relating to re-line scheduling.

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